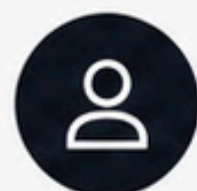




STRESSTEST

BANKING SECTOR STRESS TESTING

A scenario-based evaluation of **SBI, PNB, HDFC Bank** and **ICICI Bank** using a composite resilience framework.



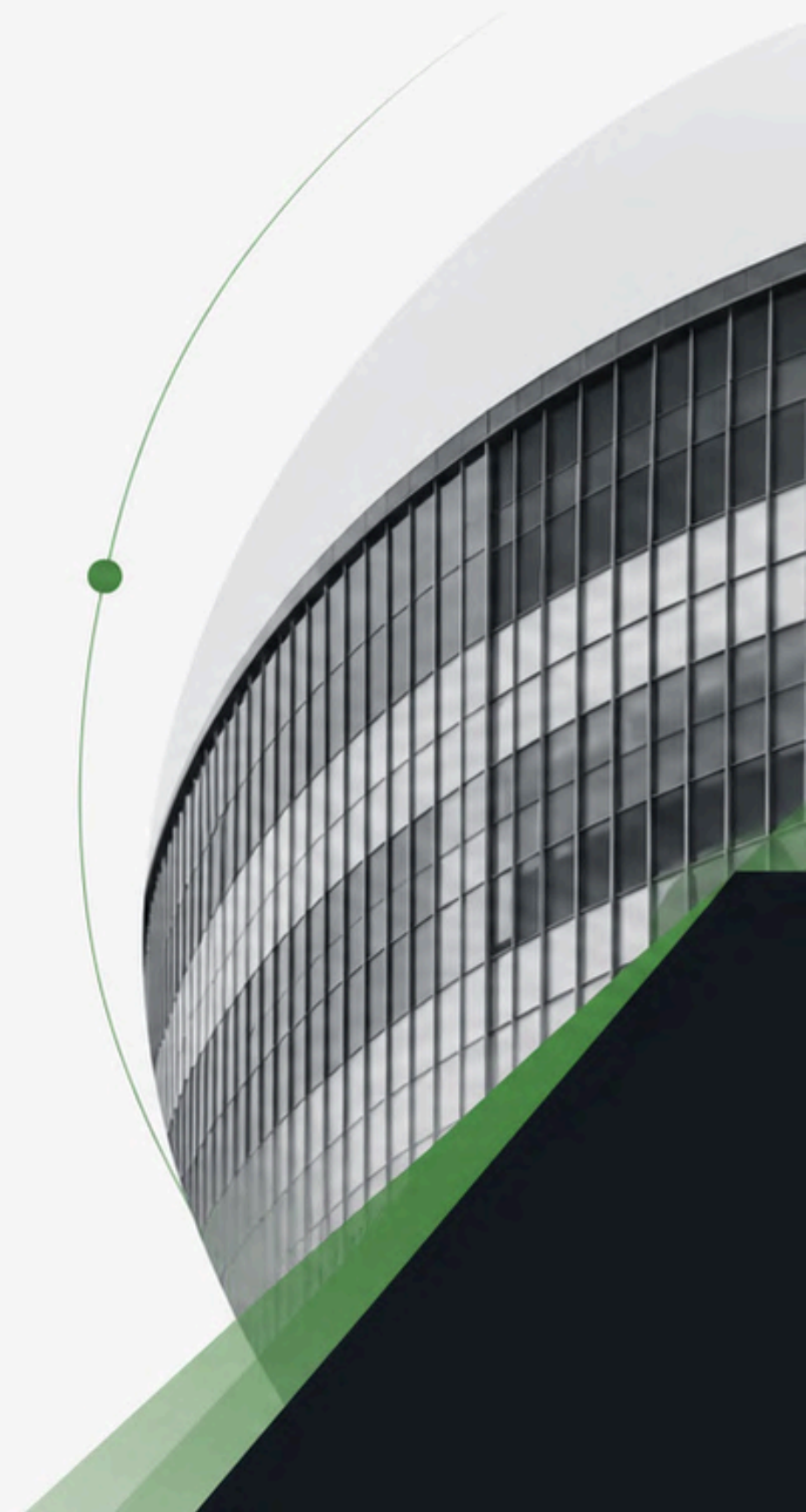
Mohit Upadhyay

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A structured walkthrough of our approach, baseline performance, stress impact and key insights from the analysis.

01		PROBLEM & OBJECTIVE Understanding the need for stress testing and the goals of this project.	04
02		METHODOLOGY & FRAMEWORK Our stress testing approach, scenarios, assumptions and BRI framework.	08
03		BASELINE RESULTS Baseline performance of SBI, PNB, HDFC Bank and ICICI Bank.	13
04		STRESS IMPACT & VULNERABILITY Impact of moderate and severe stress scenarios across key indicators.	19
05		INSIGHTS & RECOMMENDATIONS Key takeaways, PSU vs Private analysis and actionable recommendations.	28

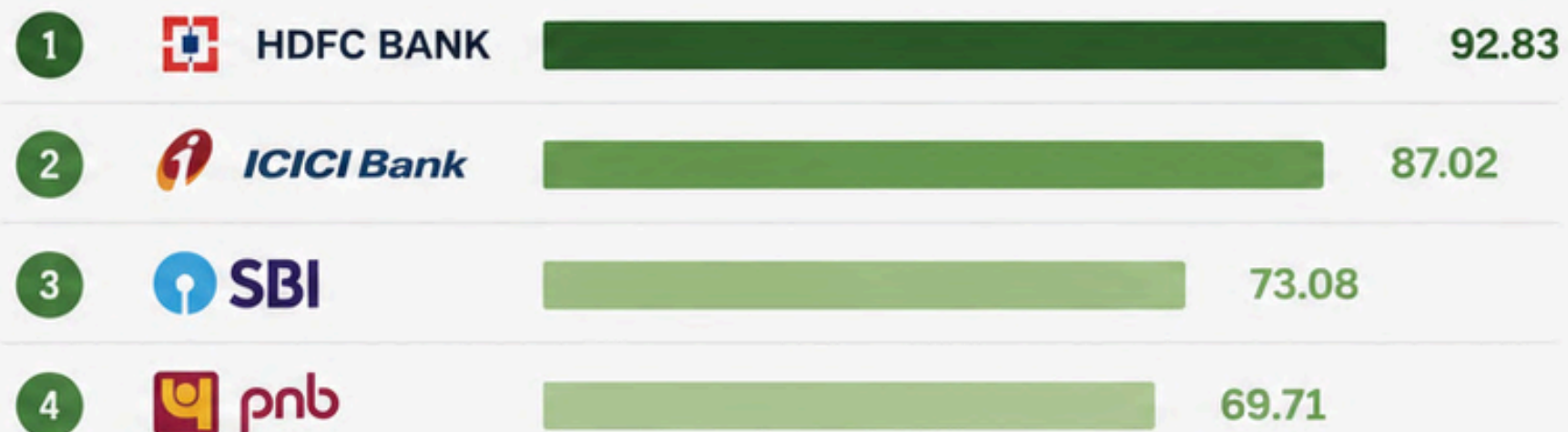


BASELINE RESULTS

WHO STARTS FROM A STRONGER POSITION?

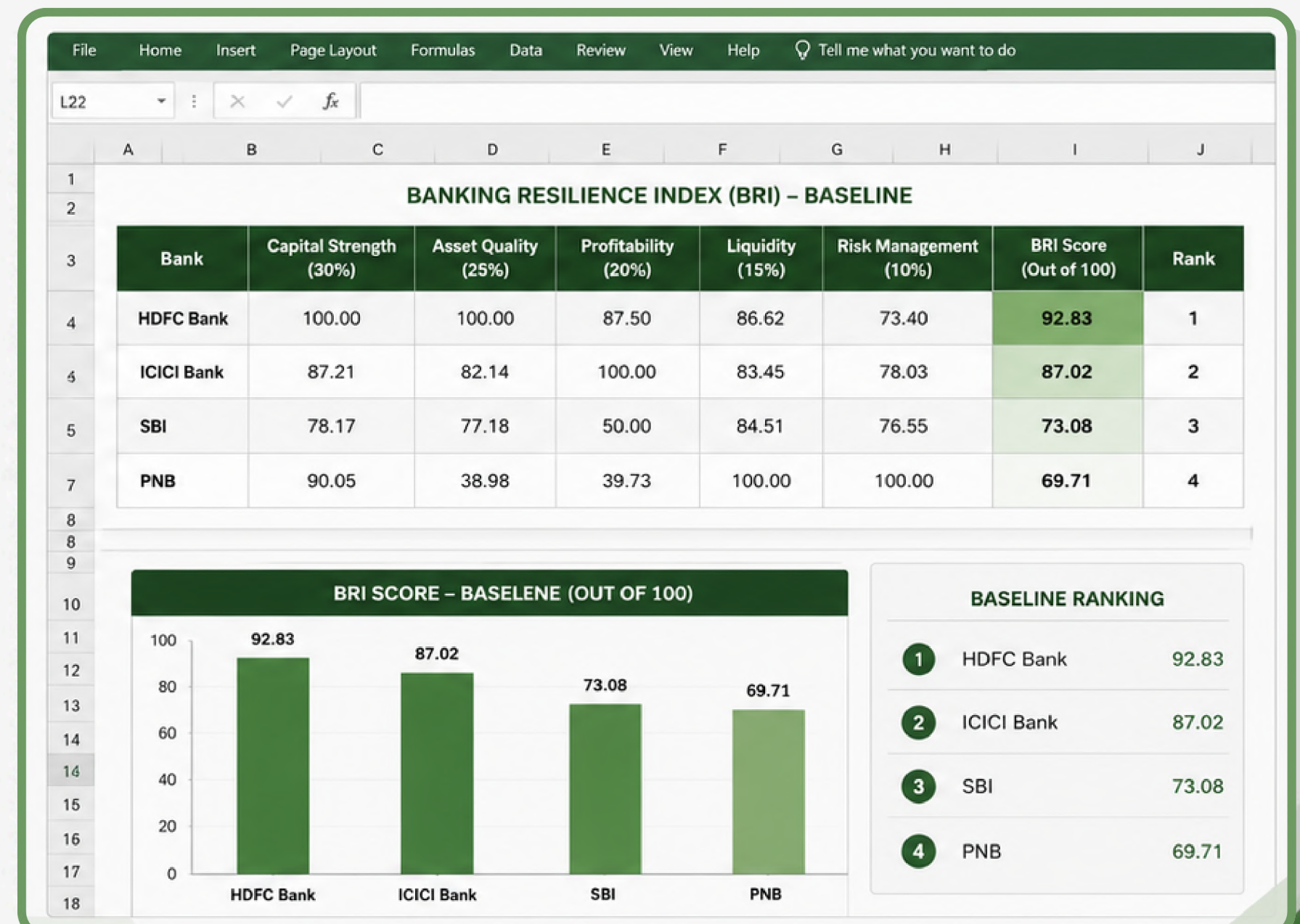
At baseline, **HDFC Bank** leads the group on overall financial resilience, followed by **ICICI Bank**. **SBI** remains structurally important but shows moderate resilience, while **PNB** is the most vulnerable among the four.

BANKING RESILIENCE INDEX (BRI) – BASELINE



BRI is a composite index (out of 100) evaluating overall financial resilience across five key dimensions.

SS of the Excel Sheet



KEY TAKEAWAY

HDFC Bank shows the strongest starting position across all components of financial resilience, while PNB reflects higher vulnerability at baseline.

STRESS IMPACT:

WHICH BANKS REMAIN RESILIENT UNDER ADVERSE CONDITIONS?



Applying moderate and severe stress assumptions reveals that **asset quality deterioration** is the primary risk transmission channel across all banks.

Private sector banks retain stronger resilience due to superior profitability and capital buffers, while **PSU banks** exhibit higher vulnerability under severe stress conditions.

KEY STRESS FINDINGS



CREDIT RISK DOMINATES

GNPA increases significantly under stress, making asset quality the largest source of financial deterioration.



PROFITABILITY COMPRESSION

ROA and ROE decline as provisioning requirements increase and earnings absorb higher credit costs.



CAPITAL BUFFERS MATTER

Banks entering stress with stronger CAR maintain better resilience and absorb losses more effectively.



LIQUIDITY REMAINS STABLE

Liquidity weakens moderately but remains less influential than asset quality and capital adequacy.

VULNERABILITY RANKING (HIGHER SCORE = MORE VULNERABLE)

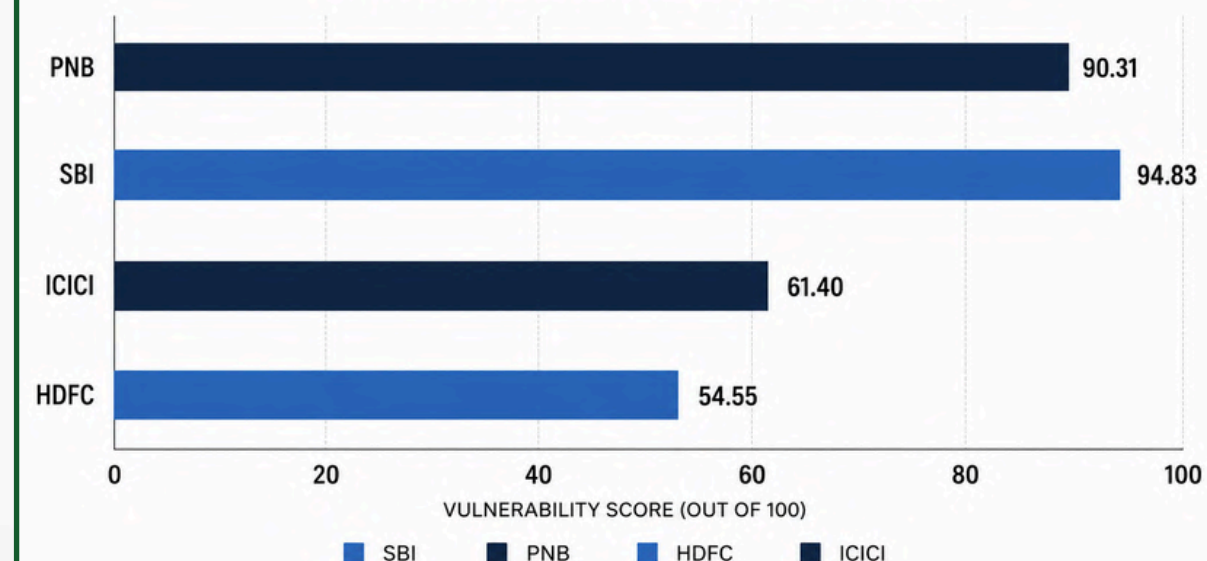


SBI and **PNB** demonstrate greater sensitivity to adverse conditions due to weaker stress performance across asset quality and profitability metrics.

HDFC Bank remains the most resilient under the severe stress scenario.

Rank	Bank	Vulnerability Score (Out of 100)
1	 SBI	76.42
2	 PNB	72.18
3	 ICICI Bank	48.35
4	 HDFC BANK	32.11

STRESS VULNERABILITY SCORE



KEY TAKEAWAY

The analysis suggests that resilience is not determined by size alone.

Banks with stronger profitability, better asset quality and healthier capital buffers absorb stress more effectively than larger institutions with weaker fundamentals.

In this framework, **HDFC Bank** and **ICICI Bank** demonstrate superior shock absorption capacity, while **SBI** and **PNB** experience greater stress deterioration.

INSIGHTS & RECOMMENDATIONS

WHAT THE MODEL IMPLIES FOR BANKS & DECISION-MAKERS?



Resilience should be judged through a **composite lens** rather than a single ratio.

A bank with high profitability but weak asset quality can still become vulnerable under stress, while a bank with **stronger buffers** can absorb shocks more effectively.

RECOMMENDATIONS BY BANK

01 SBI

- Improve asset-quality discipline and monitor stressed accounts closely.
- Maintain strong provisioning practice and reduce slippages.
- Preserve profit stability through better cost management and improved operational efficiency.

02 pnb

- Strengthen asset quality by reducing stressed exposures and improving recoveries.
- Build stronger capital buffers to improve shock absorption capacity.
- Enhance credit underwriting standards and risk controls.

03 ICICI Bank

- Balance growth with disciplined risk management.
- Sustain profitability momentum while controlling cost of risk.
- Maintain healthy capital and liquidity buffers.

04 HDFC BANK

- Protect strong liquidity position and funding stability.
- Continue maintaining superior asset-quality standards.
- Focus on long-term sustainable growth while avoiding complacency.

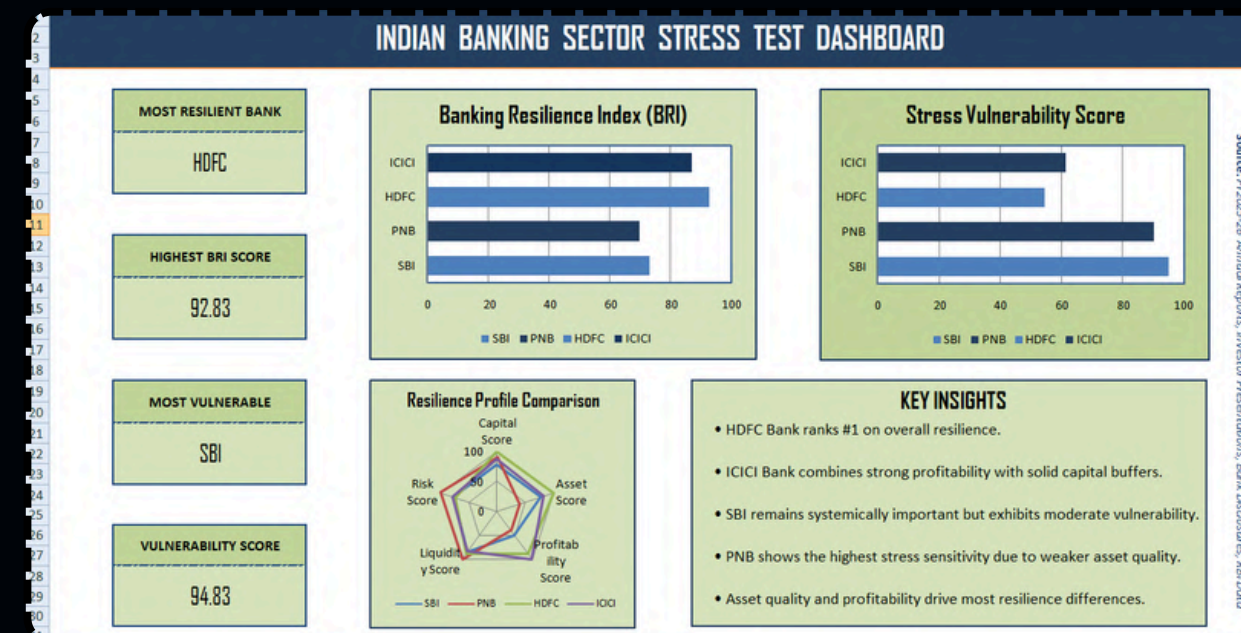


STRONG FOUNDATIONS TODAY. GREATER RESILIENCE TOMORROW.

Stress testing helps ensure financial stability, protect stakeholders and build a stronger banking system.



FINANCE &
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KEY TAKEAWAY



The project shows that stress testing is not just about identifying weak banks; it is about understanding which institutions remain stable when the environment becomes hostile.



Private sector banks in this sample demonstrate stronger overall resilience under stress due to better profitability, asset quality and capital strength.



PSU banks need focused improvements in asset quality and profitability to enhance their stress absorption capacity.

SOURCES & METHODOLOGY NOTES



DATA SOURCES USED

BANK	SOURCE
 SBI	FY2025-26 Annual Report, Investor Presentation, Q4 FY26 Earnings Release
 HDFC BANK	FY2025-26 Investor Presentation & Financial Statements
 ICICI Bank	FY2025-26 Performance Review Report
 pnb	FY2025-26 Annual Financial Statements & Earnings Call Reports



KEY METRICS USED

- GNPA Ratio
- NNPA Ratio
- Provision Coverage Ratio (PCR)
- Capital Adequacy Ratio (CAR)
- Net Interest Margin (NIM)
- Return on Assets (ROA)
- Return on Equity (ROE)
- Liquidity Coverage Ratio (LCR)



MODEL ASSUMPTIONS

MODERATE STRESS			SEVERE STRESS		
	GNPA	+1.0%		GNPA	+2.5%
	CAR	-0.5%		CAR	-1.5%
	NIM	-0.2%		NIM	-0.5%
	ROA	-15%		ROA	-40%
	ROE	-10%		ROE	-30%
	PCR	-5%		PCR	-15%
	LCR	-5%		LCR	-15%



This project is intended for educational and analytical purposes only.
Stress scenarios are hypothetical and do not represent forecasts or investment recommendations.



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